Problem set 1

1. Assume that the LU factorization $A = LU$ of a nonsingular matrix $A \in \mathbb{R}^{n\times n}$ has already been computed. Suppose that we want to solve

$$AX = B,$$

where $B \in \mathbb{R}^{n\times p}$ contains p right-hand sides.

   (a) Explain how the factorization $A = LU$ can be reused to compute all columns of X.

   (b) Determine the leading-order computational cost of solving for all p right-hand sides after the factorization is available.

   (c) Compare this with the cost of performing a new Gaussian elimination separately for every right-hand side. For what reason is a matrix factorization particularly useful when many systems with the same coefficient matrix must be solved?

   (d) Show how Gaussian elimination can be used to compute the inverse of a nonsingular matrix A. Estimate the computational cost of this procedure.

2. Show that the leading term in computation cost of the Cholesky algorithm for an $n \times n$ matrix A is

$$\frac{1}{3}n^3$$

and compare it with the LU decomposition using Gaussian elimination. Refer to Algorithm 3 in lecture notes.

3. Suppose Gaussian elimination with partial pivoting is applied to a matrix A. Using the update formula show that

$$\max_{i,j} |a_{ij}^{(k+1)}| \leq 2 \max_{i,j} |a_{ij}^{(k)}|.$$

Use this relation inductively to derive the bound

$$\rho_n \leq 2^{n-1}$$

for the growth factor. Why does this estimate not imply that Gaussian elimination with partial pivoting is unstable in practice?

4. Consider the family of linear systems

$$A_\varepsilon x = b_\varepsilon, \qquad A_\varepsilon = \begin{pmatrix} 1 & 1 \\ 1 & 1+\varepsilon \end{pmatrix}, \qquad b_\varepsilon = \begin{pmatrix} 2 \\ 2+\varepsilon \end{pmatrix},$$

where $0 < \varepsilon \ll 1$. The exact solution is $x = [1, 1]^T$. Consider instead the approximation $\widehat{x} = [0, 2]^T$. Using the infinity norm,

   (a) Compute the relative forward error.

(b) Compute the residual $r = b_\varepsilon - A_\varepsilon \widehat{x}$ and its norm.

(c) Compute the normwise backward error $\eta(\widehat{x})$ in infinity norm.

(d) Compute the condition number of A_ε in infinity norm.

(e) Explain how this example illustrates the difference between backward error and forward error.

5. Let A be symmetric positive definite with Cholesky factorization

$$A = CC^T,$$

where

$$C = \begin{pmatrix} c_{11} & & \\ * & \ddots & \\ * & * & c_{nn} \end{pmatrix}, \qquad c_{ii} > 0.$$

Define $D = \operatorname{diag}(c_{11}^2, \ldots, c_{nn}^2)$ and $L = C \operatorname{diag}\left(\frac{1}{c_{11}}, \ldots, \frac{1}{c_{nn}}\right)$.

(a) Show that L is unit lower triangular (i.e. all diagonal element of L are 1).

(b) Prove that

$$A = LDL^T.$$

(c) Deduce an LU factorization of A from this decomposition.

(d) Explain why the pivots produced by Gaussian elimination without pivoting are positive for a symmetric positive definite matrix.